

TABLE II
Gyroscope Products

Company	Product Number	Measurement Range (°/sec)	Number of sensing axes	Nonlinearity (% of Full Scale)	Shock Survivability	Bandwidth	Output Type
Analog Devices	ADXR5610	±300	Z	0.10%	2000 gm	0.01-2500Hz	Analogue
Analog Devices	ADXR5150	±150	Z	0.10%	2000 gm	14,000Hz	Analogue
Freescale Semiconductor	MMA7260Q	±1.5/2/4/6	X, Y, Z	1.00%		350/150Hz	Digital
Freescale Semiconductor	FXAS21000	±1600	X, Y, Z	1.00%		1.5625-200Hz	Digital
STMicroelectronics	A3G4250D	±245	X, Y, Z	5.00%		>140Hz	Digital
STMicroelectronics	L2G3IS	±65 / ±130	X, Y	0.10%		9.5kHz	Digital
InvenSense	ITG-31N1	±250, ±500, ±1000, ±2000	X, Y, Z		10,000 gm	400kHz	Digital
InvenSense	ITG-3200	±2000	X, Y, Z		10,000 gm	400kHz	Digital
InvenSense	IDG500	±500/110	X, Y	<1%	10,000 gm	1-1000Hz	Analogue
Maxim Integrated	MAX21000+	±31.25, ±62.5, ±125, ±250	X, Y, Z	0.20%	10,000 gm	400Hz	Digital
Murata Electronics, North America	SCR1100-D02	±100	X	±0.5%	3000 gm	50Hz	Digital/SPI
Murata Electronics, North America	SCC1300-D02 (also has accelerations sensing capability)	±100	X	±0.5%	3000 gm	50Hz	Digital/SPI
Bosch Sensortec	BMG160	±125 – ±2000	X, Y, Z	±0.05%	2000 gm, 10,000 gm	2kHz	I ² C, SPI
EPSON	XV-3500CB	±100		±5%			
MTC Industries and Research	MFG-5100	±120	1-axis	±0.2%	2000 gm		Digital
MTC Industries and Research	MRG-5200	±320	1-axis	±0.1%	2000 gm	50Hz	Analogue or digital

Rugged seems to be the keyword

It comes as no surprise though, that the sensor which is used to detect a free-fall should not be the first one to bonk out on impact.

The bias stability for gyros may be pretty numbers in the datasheet, but everything goes out of the window once you get your device out into the real world. Gravity sensitivity and environmental factors like heat, all play a part here. This is why bias stability and vibration rejections are some of the key parameters being looked at. Analog Devices' ADXR564x family of low-noise, vibration-rejecting yaw rate gyroscopes are drop-in performance upgrades to existing designs using the ADXR562x family.

For modern applications such as oil-downhole monitoring, UAV inertial measurements and industrial robotics, there is an increased demand for rugged accelerometers. Of course, another area that requires these accelerometers

is the exponentially growing mobile devices segment. With this demand in mind, the last year has seen the launch of tougher accelerometers that can withstand heavy impacts.

A new line of sensors that are insensitive to temperature changes or gradients, with signal output unaffected by electromagnetic interference, and requiring no warm-up time was brought out by Silicon Designs in the last half of 2013. All Silicon Designs' accelerometers feature a custom-integrated circuit with onboard sending amplifier and differential output stage, with a 0.5V-4.5V single-ended or ±4V differential output, proportional to the amount of measured acceleration.

Laser accelerometers

Researchers at Caltech, the California University of Technology, are working on accelerometers that work with lasers. An accelerometer normally uses an electrical circuit, whilst a laser accelerometer uses laser light instead

of electricity. The optical cavity of this accelerometer is very small (about 20 microns long, only a single micron deep and few tenths of a micron thick). It contains two silicon nanobeams that are situated in an accelerometer as the two sides of a zipper that has a proof mass attached to one of its sides.

The moment a laser light enters the accelerometer, the nanobeams act as a 'light pipe'. These nanobeams guide the light to be bounced back and forth in between its holes. The movement of the proof mass results in the change of the intensity of the laser light that is reflected out. This reflected laser light is so sensitive to the motion of the proof mass that it helps in the determination of even the slightest displacement.

The laser accelerometers are still under research to find the cost-effective ways of using laser with accelerometers. ●

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